Capture Systems Embedded Software Engineer Take-Home Assignment

We're looking for an embedded software engineer to join our capture team to build the software that reliably gets data from our cameras to the cloud.

Problem Statement
You are given a capture rig with two pan-tilt cameras on Ethernet. Each camera streams raw video at 125 Mb/s, and its position 200 times a second, to a small Linux computer next to it: the server. The server serves a web page where an operator watches and steers the cameras. When the operator starts a recording, the server records every frame from both cameras, with the time it was taken and where its camera was pointing, until the operator stops it. It keeps the recordings safe and uploads them to cloud storage over a 250 Mb/s link. You may compress them.

Your goal is to **build the server software in C++26 on Linux**, and connect our web viewer to it.

The server is a small computer: your software gets 1 CPU core, 250 MB of memory and 2 GB of disk, and we enforce these limits when we run it. Expect everything around it to be unreliable: the cameras won't wait for your server, connections drop, and the power can fail. Your server can't keep every frame through all of this, so it should make it clear what was lost. What it keeps, how it behaves when things go wrong, and what an operator needs to see are up to you.

Deliverables
For this take-home, your final deliverable is a GitHub repository or a zip with:

- **the source code of your server**, with a README that tells us how to build it, how to run it with the camera simulator, the cloud storage simulator and the viewer, and how to test it
- **the viewer, connected to and served by your server.** The viewer must show your server's live state, previews, data pipeline and recordings, and its controls must work
- **automated tests** for the code you write, which can include the viewer, run by GitHub Actions if you send a GitHub repository, or by one script (`./test.sh`) if you send a zip
- **a few slides or a short document with your design decisions and your system architecture**

You may use AI tools to help you build your solution.

We provide a camera simulator that acts as the two pan-tilt cameras on your own computer, a cloud storage simulator, the web viewer (React + TypeScript, complete except for the connection to your server), example data recorded from one of the cameras, and a README.md that documents the data formats. No hardware is needed. We expect you to understand and parse the formats provided.

Our goal is to understand your design process. We want to understand how you approached the problem and the trade-offs you made.

Apart from the contents of this assignment, we'll also ask you how you would handle deployment, updates and rollbacks of this software on multiple servers at remote sites with poor networks.

This is a **timed take-home.** You'll have 24 hours to complete this assignment, starting when the initial email is sent.

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